

[Paper Link](#)

1. What is the main hypothesis of the paper regarding MLPs and in-context learning (ICL), and how does it challenge traditional Transformer-based approaches?

The paper's core claim is that ICL is not unique to Transformers. Even plain MLPs and MLP-Mixers can learn in-context on standard synthetic ICL benchmarks (regression + classification) without attention. It challenges the Transformer story because most prior explanations of ICL assume something like ICL works because attention implements a learning algorithm in-context. Also the paper mentions that we don't need attention for the phenomenon of ICL-MLPs that can approximate the same computation on these tasks. So the "attention is the algorithm" narrative is not necessary to explain ICL on common toy setups.

2. Compare the performance of MLPs vs. Transformers across the toy tasks. What parameter efficiency gains does the paper report, and under what conditions do MLPs outperform Transformers?

MLPs, MLP-Mixers, and Transformers provide near-Bayes-optimal performance on toy in-context regression and classification tasks when normalized by training compute. With Transformers having just a slight advantage at extremely low compute. According to the research, MLPs may be comparably parameter-efficient under equivalent compute budgets and on some relational reasoning and ICL-like classification tasks on the other hand MLPs even exceed Transformers in terms of compute efficiency and out-of-distribution generalization. MLPs outperform Transformers when contexts are short to moderate and task structures are linear or similarity-based, but Transformers maintain their advantages as context length increases.

3. What are the key limitations identified in the paper for scaling MLPs to real-world ICL (e.g., longer contexts or complex data)? How does the authors' "linear attention hypothesis" address these?

The main limitations for scaling MLP-based ICL to real-world settings are poor handling of long contexts, rapid parameter growth due to flattened inputs and weaker inductive bias for sequence structure compared to Transformers. The author's linear attention hypothesis explains these gaps by arguing that many ICL tasks reduce to linear attention or kernel regression over context examples, which Transformers implement efficiently and stably for long sequences, while MLPs must approximate the same computation with far less structural efficiency, making scaling difficult.

4. Explain how the paper approximates linear attention using MLPs. What mathematical insight enables MLPs to perform ICL competitively with Transformers on these tasks?

The paper shows that linear attention can be written as a weighted linear combination of context labels based on similarities between the query and context points, equivalent to kernel regression or ridge regression in closed form. MLPs can approximate this same low-rank linear operator by learning nonlinear feature maps and interaction terms over the flattened context, enabling them to behave like linear attention on synthetic ICL tasks. The key mathematical insight is that ICL in these settings reduces to learning a linear estimator over context features, which an MLP can approximate given enough width, depth and data diversity.

5. How would you modify MLPs for GLUE benchmarks?

To adapt MLPs for benchmarks like GLUE, one would need to introduce token-mixing layers (as in MLP-Mixers), low-rank or pooled representations of long contexts to avoid quadratic parameter growth and possibly explicit similarity or relational features to mimic the kernel structure of linear attention. Moreover, MLPs would likely require more diverse pretraining data and careful curriculum design than Transformers to reliably transition from in-weight learning to true in-context learning on complex natural language tasks.